Lab: Descriptive statistics

Estimated reading time: 30 minutes

In this lab, you'll learn to configure Data Analysis ToolPak. You'll also be able to explore how to use Descriptive Statistics in Excel. You'll also be able to interpret the results of descriptive statistical analysis.

Software Used in this Lab

The instruction videos in this course use the full Excel Desktop version, as this has all the available product features. However, for the hands-on labs, we will use the free 'Excel for the web' version as this is available to everyone.

Although you can use the Excel Desktop software if you have access to this version, it is recommended that you use Excel for the web for the hands-on labs, as the lab instructions specifically refer to this version. There are some small differences in the interface and available features.

Data Set Used in this Lab

The data set used in this lab is curated from the following source: https://www.kaggle.com/datasets/yasserh/housing-prices-dataset under the **CC0: Public Domain license**.

Acknowledgment and thanks also go to https://trak.in, who were generous enough to share the data publicly for free.

We are using a modified subset of that data set for the lab. To follow the lab instructions successfully, please use the data set provided in this lab rather than the data set from the original source.

Learning Objectives:

- Configure Data Analysis ToolPak
- Use the Descriptive Statistics function in Excel to analyze a data set
- Interpret results of descriptive statistical analysis

Welcome to the lab: Descriptive Statistics!

Exercise: Descriptive Statistics

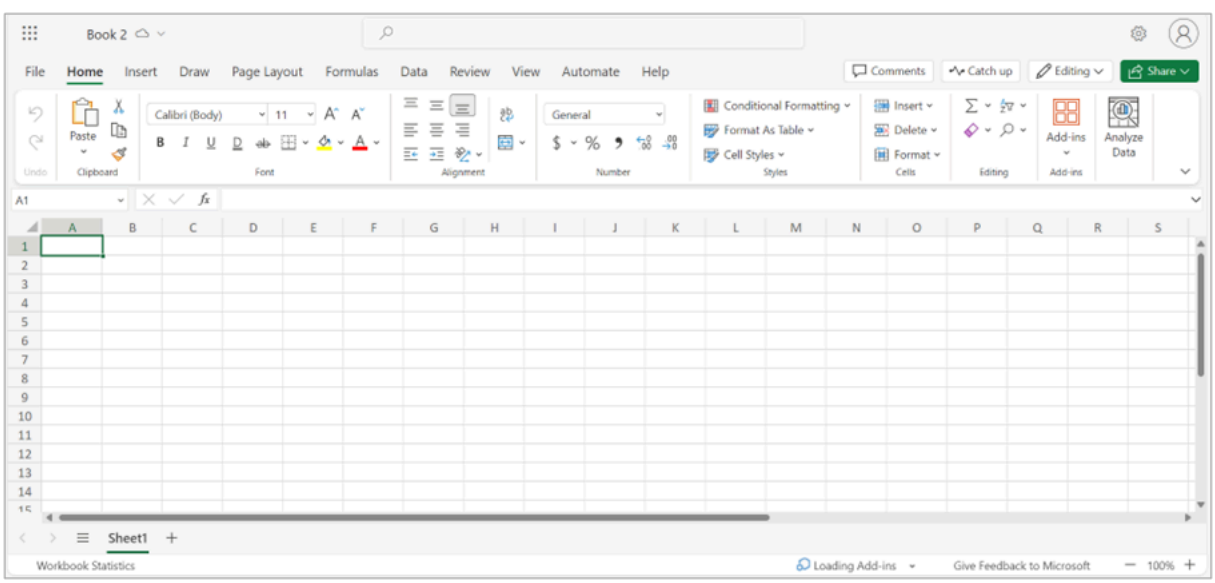
In this exercise, you'll configure the add-in XLMiner Analysis ToolPak available in the Excel for the web. You'll also measure central tendency, dispersion, skewness, and kurtosis using Descriptive statistics.

Note: XLMiner Analysis ToolPak is a third-party free statistical analysis ToolPak in Excel for the web. It shows the same functions as the Analysis ToolPak add-in available in Excel for the desktop. You can use this ToolPak to perform Descriptive Statistics and Linear Regression for the labs in this course. If you are using Excel for the desktop, you can configure in-built Data Analysis ToolPak available from Microsoft.

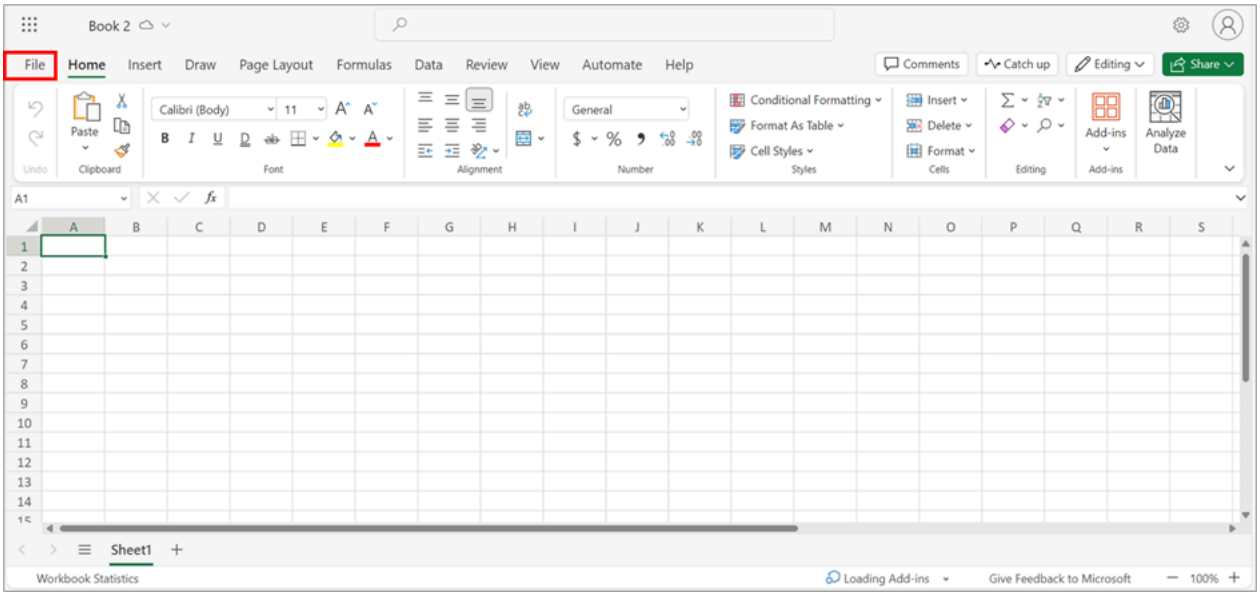
Task 1: Configure XLMiner Analysis ToolPak

In this task, you'll configure XLMiner Analysis ToolPak in Excel for the web. To do so, log in to Excel for the web and perform the steps below.

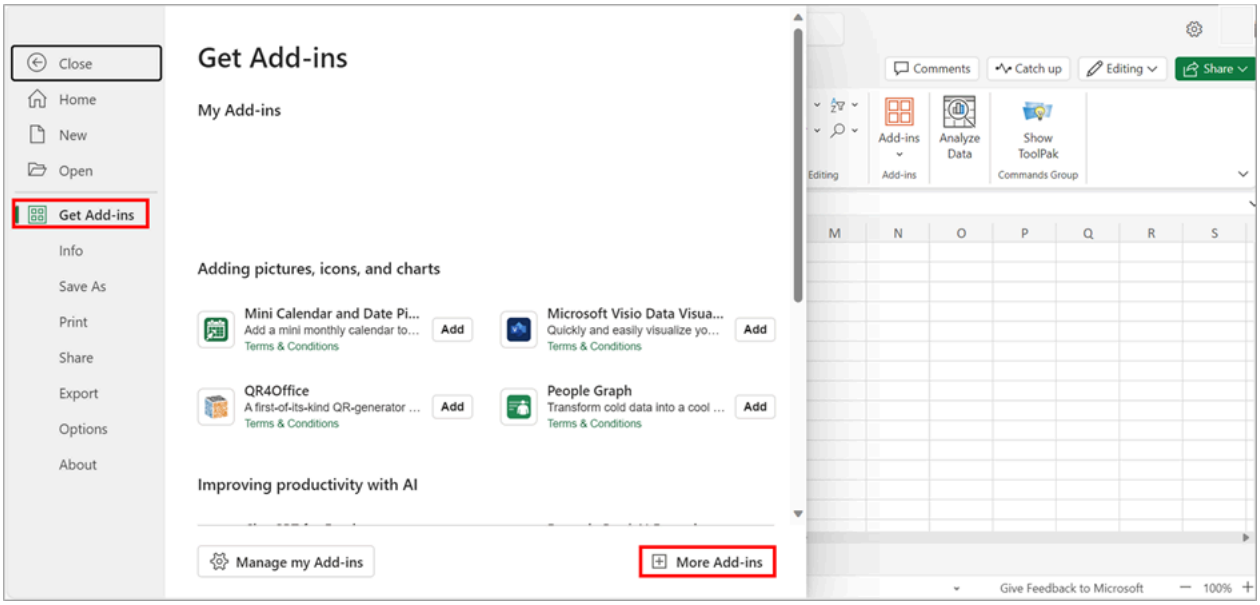
1. Open Excel for the web on your computer or laptop.



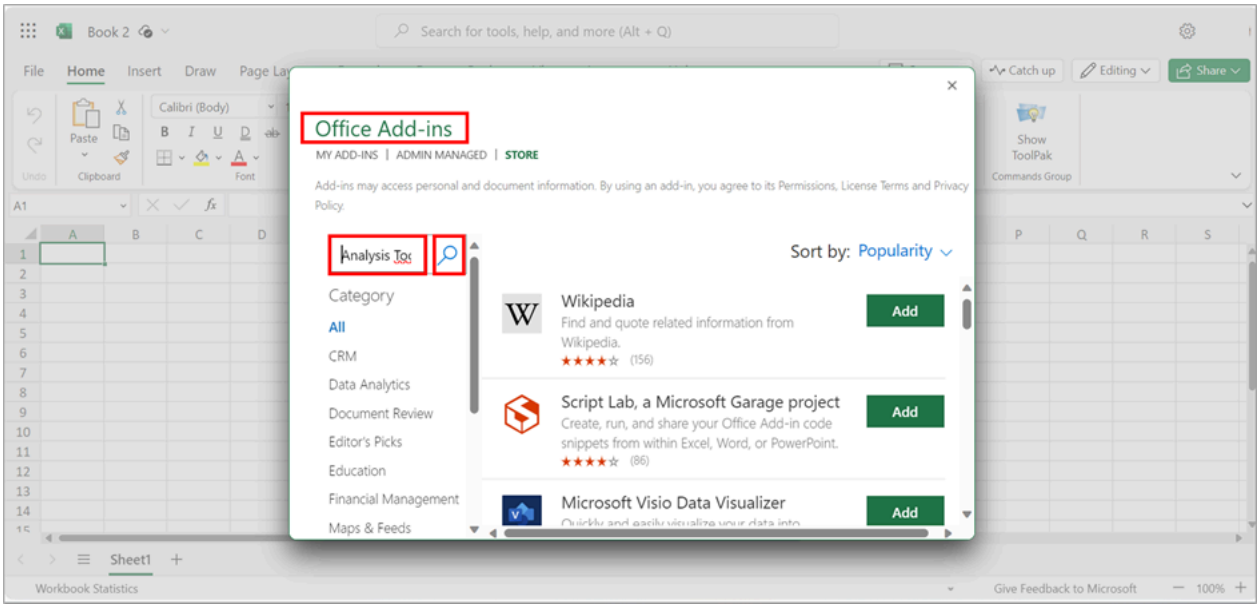
2. Select **File** on the top ribbon.



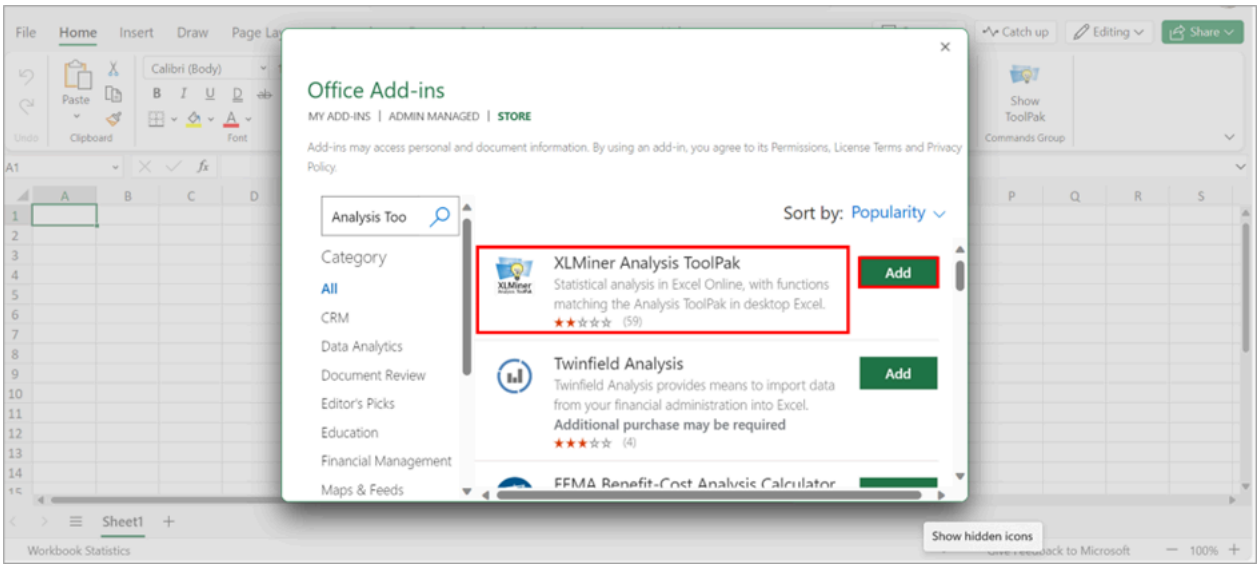
3. Select **Get Add-ins** from the left menu bar and choose **More Add-ins**.



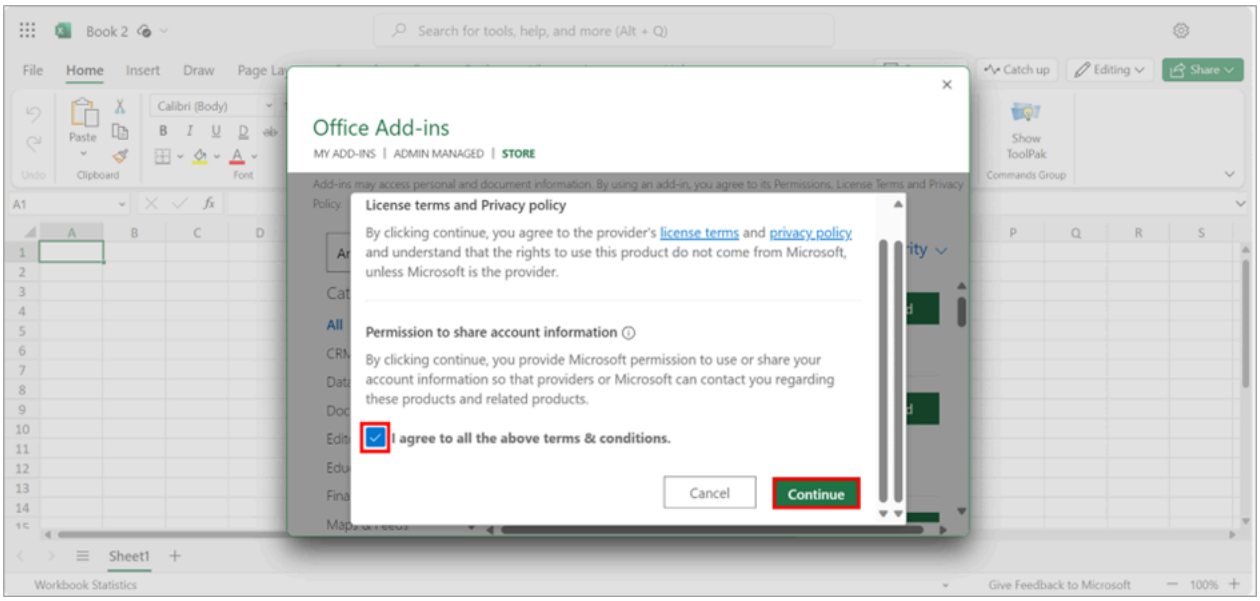
4. A window **Office Add-ins** will appear. In the **Search** section, type in the keyword **Analysis ToolPak** and select the search icon.



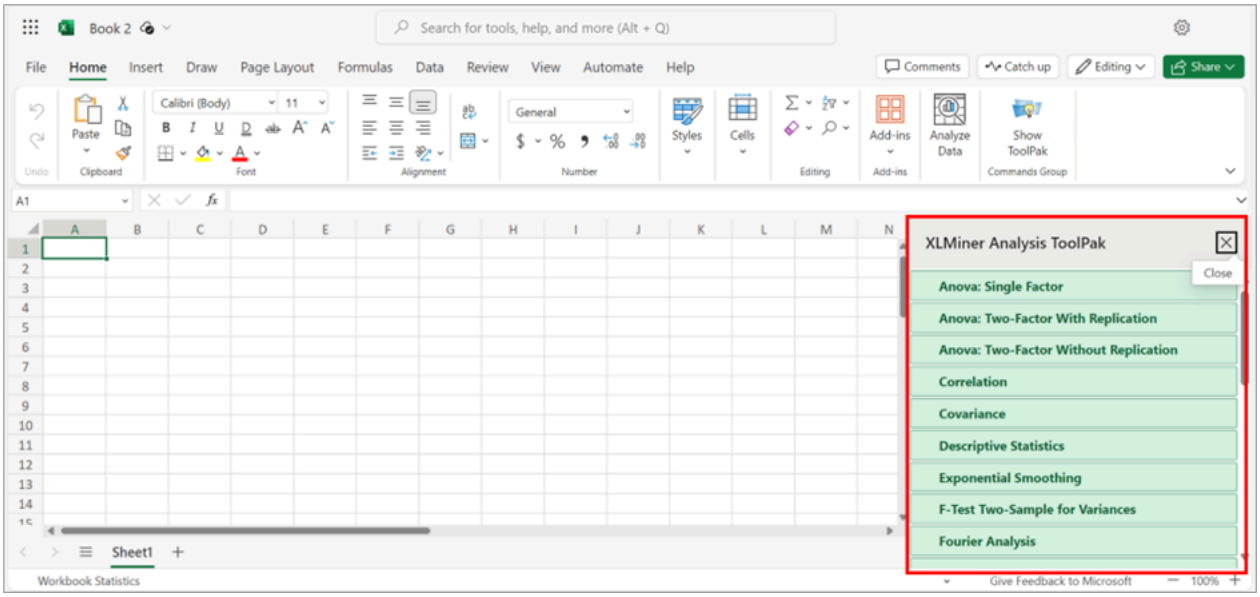
5. An **XLMiner Analysis ToolPak** will appear. Select **Add**.



6. Checkmark **I agree to all the above terms & conditions** for license terms and privacy policy. Select **Continue**.



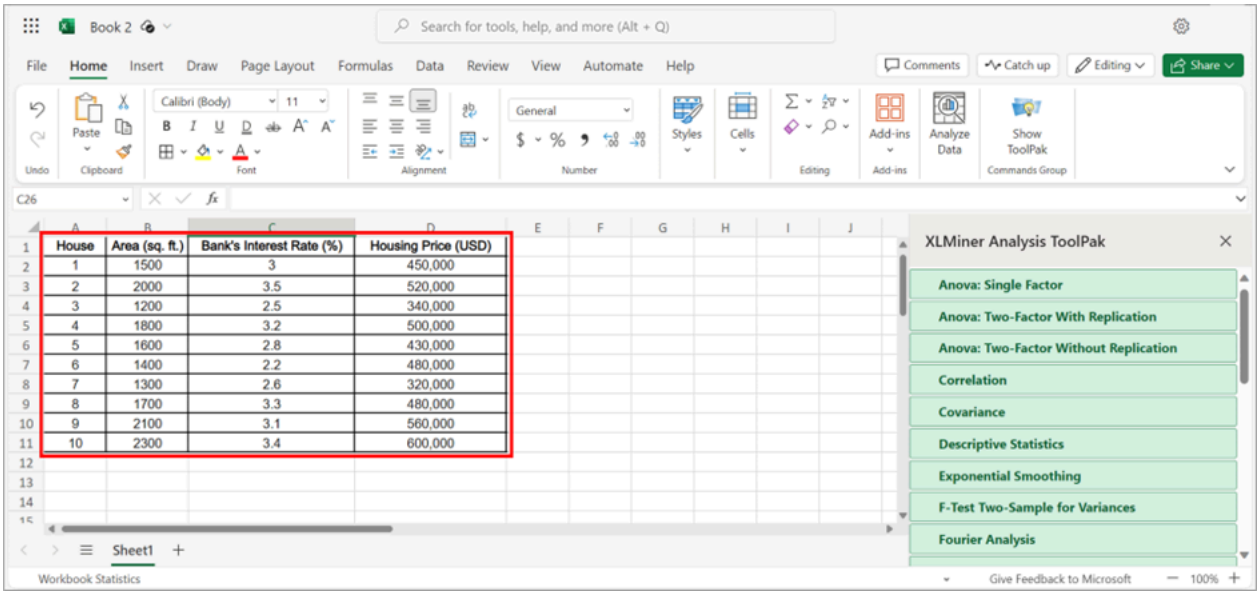
7. You'll see **XLMiner Analysis ToolPak** on the right side of your Excel sheet.



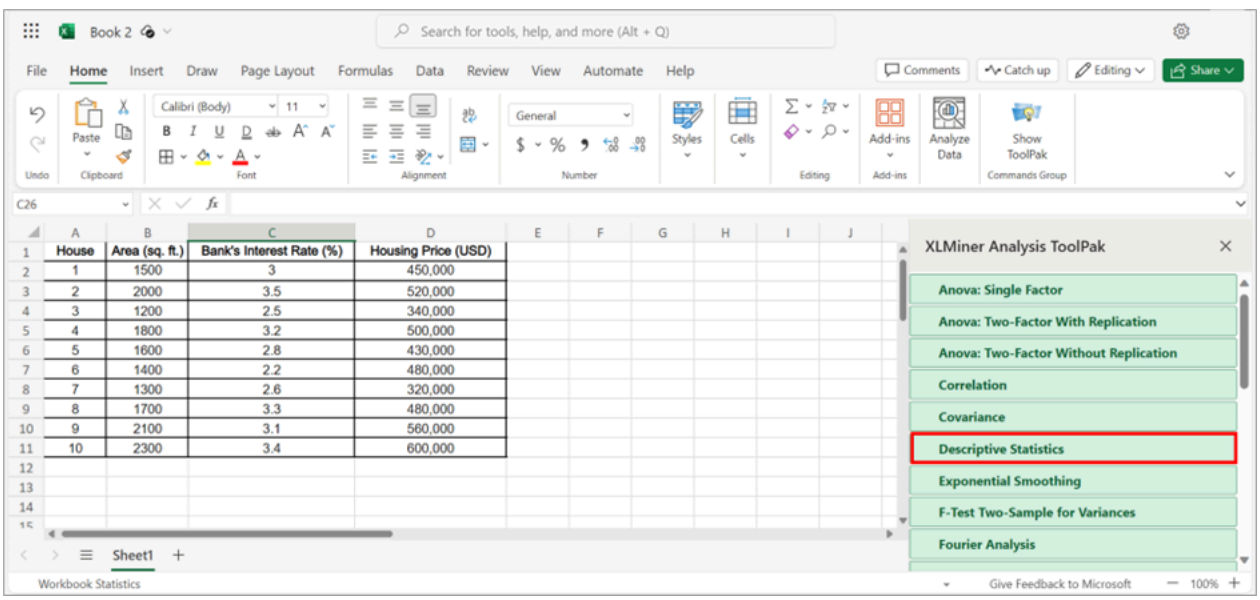
Task 2: Calculate Descriptive Statistics Using Excel

In this task, you'll learn to calculate a measure of central tendency and dispersion using Descriptive Statistics using Excel. You'll also be able to calculate and interpret skewness and kurtosis. Follow the below steps to perform these calculations.

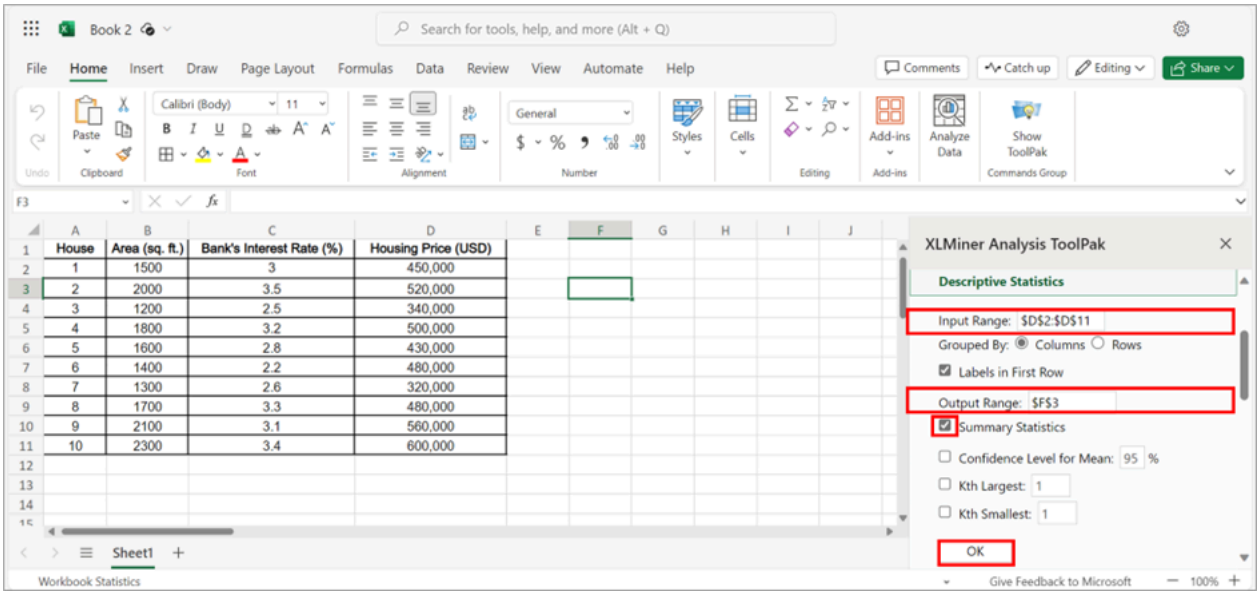
1. Copy and paste the data set from the source data into an Excel worksheet. In this case, you can use the data set for the housing prices (To download the given data set in this lab, right click on the given hyperlink and select Open link in new tab).



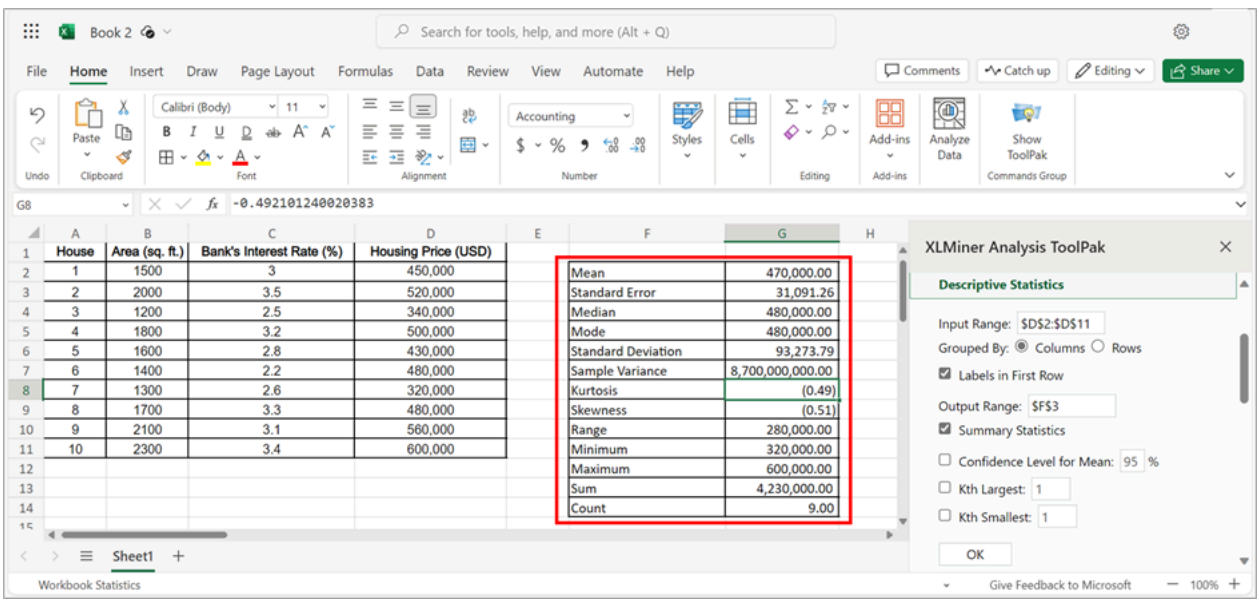
2. Select **Descriptive Statistics** from the **XLMiner Analysis ToolPak**.



3. In the **Descriptive Statistics** section, select **Input Range** as $D$2:$D$11 and **Output Range** as $F$3. Select the **Summary Statistics** checkmark. Select **OK**.



4. All the descriptive statistics for the given data set will appear in Excel for the web. You can see that the computed mean for the given score is 470,000, the standard error is 31,091.26, and the median and mode are 480,000, respectively. The standard deviation is 93,273.79, the sample variance is 8,700,000,000, the kurtosis is (0.49), the skewness is (0.51), the range is 280,000, the minimum score is 320,000, the maximum score is 600,000, the sum is 42,30,0000, and the count is 9.



Congratulations! You've completed the lab Descriptive Statistics. Now, you are ready to move to the next topic.

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